

Module 4: Individual Condition Expectation Plots And Partial Dependence Plots

Instructional Hours: 6

Module Overview

This module will provide you with a comprehensive introduction to Partial Dependence Plots (PDP) and Individual Conditional Expectation (ICE) as essential tools for model interpretability in explainable AI. You will explore the key concepts, differences between global and local explainability, the practical implementation of PDP and ICE, and how these methods help interpret complex machine learning models. The module will also introduce the strengths, limitations, and real-world applications of both techniques.

Learning Outcomes

At the end of this module, you will be able to:

1. Explain how Partial Dependence Plots (PDPs) and Individual Conditional Expectation (ICE) Plots are Used in XI
2. Discuss the process of creating PDPs and ICE
3. Create code to implement PDP and ICE using python
4. Interpret PDPs and ICE plots of AI models to stakeholders

Introduction To Individual Condition Expectation (Ice) Plots And Partial Dependence Plots

ICE Plots are a local explanation method that shows a visualization of how changing a specific feature affects the predicted outcome for a given data point, keeping all other feature values constant. It shows a separate line for each instance in the dataset, allowing us to observe individual-level feature effects. They are a local interpretability technique. A local interpretability technique explains the model behavior for an individual data instance. As an example, in figure 3.1, each blue line corresponds to one instances in the data set. For a specific instance, the value of land size is varied from 400 to 1,400 m² and the output of the model (the house price) is observed. The land size is then plotted against the price for that instance. The same is repeated for all the other instances (hence the multiple blue lines).

The orange line in Figure 3.1 shows the changes in the average house prices for all instances as the land size is varied. This is the Partial Dependence Plot

which is a plot of the feature values (smallest to largest) against the average predicted value for all instances in the dataset. The partial Dependence plot is a global interpretability technique because it explains the behavior of the model over the entire dataset.

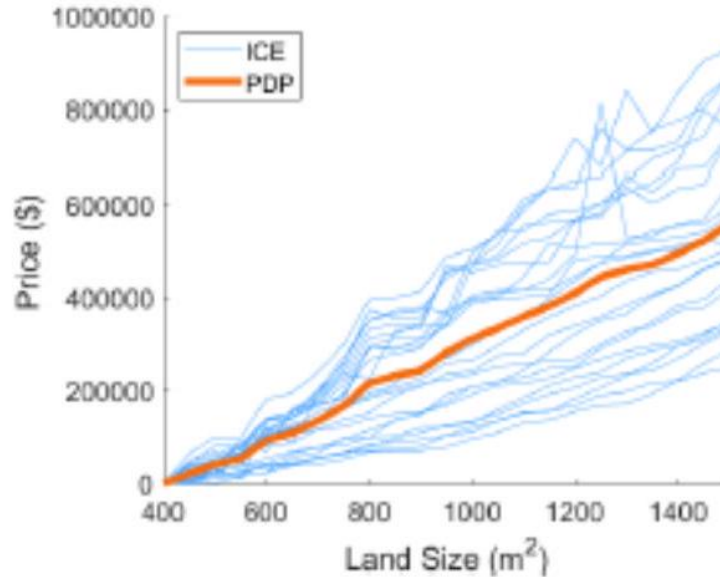


Figure 3.1: A partial Dependence plot (orange line) of Land Price against House price. As the land size increases the price also rises (Baker S. & Xiang S., 2023).

General procedure for ICE

To create ICE plots, follow the following sequence of steps:

1. Train a machine learning model (e.g., a random forest regressor) on a dataset.
2. Select a feature of interest land size.
3. For each instance in the dataset, vary the values of that feature from minimum to maximum and record the predicted outcomes.
4. Plot these outcomes to visualize how different instances respond to changes in the feature.

General Procedure for PDP

To create a PDP plot, follow the following sequence of steps:

1. Step 1: Train a machine learning model (e.g., decision tree, random forest) on a dataset.
2. Step 2: Choose a feature of interest (e.g., land size in a house price prediction model).
3. Step 3: For each value of that feature, compute the average model prediction across the dataset, while keeping all other features constant. Step 4: Plot these average predictions to visualize the relationship between the feature and the target.
4. Step 4: Plot these average predictions to visualize the relationship between the feature and the target.

Interpreting ICE Plots

Instance-Level Insights

Each line in the ICE plot represents how a specific instance's prediction changes with variations in the feature. This helps in understanding individual predictions rather than relying on aggregate patterns. It can help answer the questions: How is this instance affected by a change in this feature? What would have been the output for this instance if this feature was set to this specific value?

Heterogeneous Effects

ICE plots highlight whether different instances are affected similarly or differently by the feature. For example, in some cases, the model's predictions may increase for one group while decreasing for another.

Feature Interaction

ICE plots help detect interactions between features, which might not be visible in PDPs. If the lines in the ICE plot behave differently for different data points, it suggests that the feature's effect on prediction may depend on other features.

Interpreting Partial Dependence Plots

Positive/Negative Slope

If the PDP curve has a positive slope, it indicates that the feature is positively correlated with the predicted target. A negative slope suggests the opposite.

Non-linear Relationships

A non-linear curve suggests a complex relationship between the feature and the target. For instance, predictions might increase for a certain range of feature values, but plateau or decrease after a threshold.

Feature Interactions

PDPs can also reveal interactions between two features when plotted together. For example, a 2D PDP plot (with two features) shows how the interaction between these features affects the model's predictions.

Strengths and Limitations of ICE Plots

Strengths

Local Interpretability

ICE provides detailed, instance-level insights, which are particularly useful in domains where understanding individual behavior is critical (for example health-care, finance).

Detecting Heterogeneous Effects

ICE helps in identifying whether the effect of a feature varies across different data points.

Complementary to PDP

ICE is often used alongside PDP to provide a more complete understanding of feature effects, allowing both global and local interpretation.

Limitations

Complexity of Interpretation

ICE plots can be overwhelming, especially when there are many data points, making it difficult to extract clear patterns.

Computational Expense

Generating ICE plots for large datasets or highly complex models can be computationally intensive.

Strengths and Limitations of PDP

Strengths

Global Interpretation

PDP offers a global perspective on how a feature influences the model's predictions, making it valuable for understanding feature importance at a high level.

Simplicity

PDPs are easy to compute and interpret, particularly for linear models or models where the effect of a single feature dominates.

Model-Agnostic

PDP can be applied to any machine learning model, from linear models to complex ensemble methods like random forests or gradient-boosted machines.

Limitations

Assumption of Feature Independence

PDP assumes that features are independent of one another, which is rarely true in real-world data. When features are correlated, the PDP may not accurately reflect the true effect of the feature.

Limited to Global Insights

PDP provides a global interpretation, meaning it shows the average effect of a feature across the entire dataset. It does not offer local explanations (i.e., it does not explain individual predictions). For local interpretation, Individual Conditional Expectation (ICE) should be used.

Computational Cost

For large datasets or complex models, generating PDPs can be computationally expensive.

Detailed Review Of Pdp And Ice Plots Examples

Reading Material

To learn about Individual ICE Plots and Partial Dependence Plots, read the following materials.

Individual Conditional Expectation

Molnar, C. (2022). Interpretable Machine Learning: A Guide for Making Black Box Models Explainable (2nd ed.), URL: <https://christophm.github.io/interpretable-ml-book/ice.html>

Partial Dependence Plots

Molnar, C. (2022). Interpretable Machine Learning: A Guide for Making Black Box Models Explainable (2nd ed.), URL: <https://christophm.github.io/interpretable-ml-book/pdp.html>

Implementation Of Ice Plots And Pdp Using Python

You can implement ICE plots and PDP using python. You should use the `PartialDependenceDisplay.from_estimator` class from the `sklearn.inspection` package. In the code below, we use the permutation features importance to determine the importance of features to enable us select which feature to plot in the ICE plots and PDP. We will use the `breast_cancer` dataset from the `sklearn` datasets.

```

from sklearn.ensemble import RandomForestClassifier as RFC
from sklearn.metrics import accuracy_score
from sklearn.inspection import PartialDependenceDisplay
from sklearn.inspection import permutation_importance
from sklearn.model_selection import train_test_split
from sklearn.datasets import load_breast_cancer
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
#Load the data
data=load_breast_cancer()
#print(data.DESCR)
#Check the dimensions of the data
x=data.data
print(x.shape)

y=data.target
print(y.shape)
#Get the columns of the data
columns=data.feature_names.copy()
print(columns)

import pandas as pd
df = pd.DataFrame(x, columns=columns)
df['target'] = y

#Put the data in a dataframe for easier visualization

#change benign to zero and malignant to 1
df['target']=df['target'].apply(lambda x: 1 if x==0 else 0)
df.head()

# split the data into training and test set
x=df[columns]
y=df['target']
x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.4, random_state=42)

# fit the model
model=RFC(n_estimators=100, random_state=42)
model = model.fit(x_train, y_train)

#print(model.summary())

```

```

importances=[]
for col, val in zip(columns,model.feature_importances_):
    importances.append((col, val))
importances.sort(key=lambda x: x[1], reverse=True)
print(importances)

#perform some predictions
predictions=model.predict(x_test)
for val in predictions[0:10]:
    print("{:.0f}\t".format(val), end="")
print()
for val in y_test[0:10]:
    print("{:.0f}\t".format(val), end="")
print()
acc=accuracy_score(y_test, predictions)
print("Accuracy: {:.2f}".format(acc))

# Apply Permutation Feature Importance
result = permutation_importance(model, x_test, y_test, n_repeats=10, random_state=42, scoring='accuracy')
dfSummary=pd.DataFrame({'Variable':columns, 'importance':result.importances_mean, 'coeff_lb':result.importances_lower_bound, 'coeff_ub':result.importances_upper_bound})
#Sort the values
dfSorted=dfSummary.sort_values(by="importance", ascending=False)
dfSorted.head(30)

#plot the ICE plots and PDP
PartialDependenceDisplay.from_estimator(model, x_test, ["worst area"],feature_names=columns,
plt.show()

```

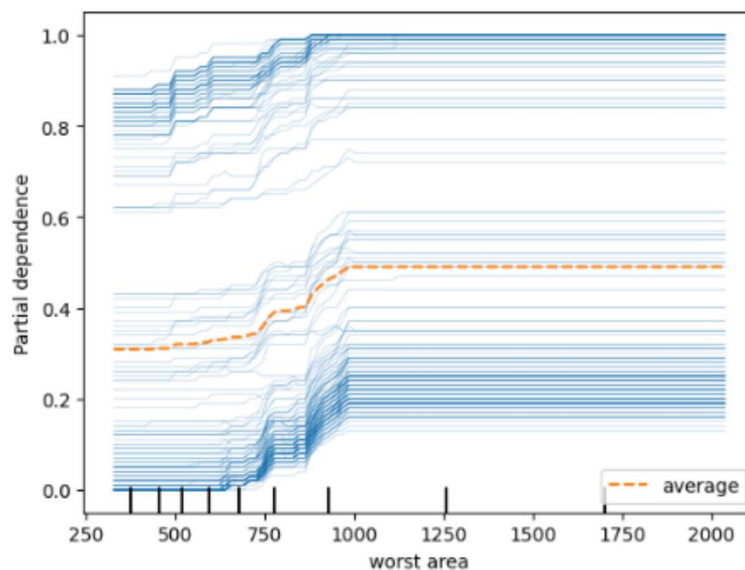


Figure 3.2: ICE Plot and PDP of the worst area feature from the scikit-learn breast cancer data set.

From Figure 3.2, you can see that there are two clusters in the data set. From the top cluster, an increase in worst area from about 400 to 1,000 results in an increase in probability of malignancy in the breast mass sample. In the lower cluster, an increase in worst area from about 600 to 1,100 leads to a slight increase in the probability of malignancy. The PDP also seems to reflect this increase up to about 1,100.